KAPILAN BALAGOPALAN

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INTRODUCTION

My research focuses on sequential decision-making, including bandit algorithms, partial monitoring games, and reinforcement learning with applications in fine-tuning and improving large language models and recommender systems. I have a strong foundation in theoretical machine learning and foundational models, complemented by hands-on experience in embedded C, C++, Python, Java, MATLAB, and API-level development for foundation models.

EDUCATION

University of Arizona, USA *Aug 2022 - present*
PhD in Computer Science
Overall GPA: 3.67/4.0

University of Moratuwa, Sri Lanka *Feb 2014 - Jan 2018*
B.S.(Hons) in Engineering
Department Electronic and Telecommunication Engineering
Overall GPA: 3.81/4.2

PROFESSIONAL EXPERIENCE

Amazon May 2025 - August 2025
Applied Scientist Intern *Amazon, New York*
Campaign intent detection using large language models and using the intents to recommend key-words for advertisers to create optimal campaigns.

Thales Digital Identity & Security May 2020 - May 2022
Embedded Software Engineer *Thales DIS, Singapore*
Development of Smart card OS for Telco, Secure Element, Transport and Banking applications to be deployed on micro controller chips.

London Stock Exchange Group Feb 2018 - Feb 2020
Hardware Acceleration Engineer *LSEG Technology, Sri Lanka*
Acceleration of time critical modules in exchange and trading platforms using FPGAs.

Temasek Laboratory, Nanyang Technological University Aug 2016 - Dec 2016
Research Assistant *NTU, Singapore*
Blind channel equalization of OQPSK signals under strong co-channel interference using higher order statistics.

TEACHING & VOLUNTEERING ROLES

- I have been a reviewer in top-tier conferences like ICML, Neurips, ICLR and AISTATS.
- Teaching Assistant
  - Spring 2025: CSC 445 Algorithms
  - Fall 2025: CSC 477/577 Computer vision

JOURNALS & CONFERENCES

† indicates equal contribution.

- **NeurIPS'26**(*under review*): ***Efficient Algorithms for Contextual Apple Tasting with Log-Loss.*** Byeongwoo An†, **Kapilan Balagopalan**†, Sehwa Jeong, Jiun Jeong, Kyoungseok Jang, Hyowon Wi, Noseong Park, Gi-Soo Kim, Kwang-Sung Jun. Advances in Neural Information Processing Systems. 2026.
- **ICML'26**: ***Fixed Budget is No Harder Than Fixed Confidence in Best-Arm Identification.*** **Kapilan Balagopalan**†, Yinan Li†, Yao Zhao, Tuan Nguyen, Anton Daitche, Houssam Nassif, Kwang-Sung Jun. International Conference on Machine Learning. 2026.
- **ICML'25**: ***Fixing the Loose Brake: Exponential Tail Bounds for Stopping Time in Best Arm Identification.*** **Kapilan Balagopalan**†, Tuan Nguyen†, Yao Zhao†, Kwang-Sung Jun. International Conference on Machine Learning. 2025.
- **AISTATS'25**: ***Minimum Empirical Divergence for Sub-Gaussian Linear Bandits.*** **Kapilan Balagopalan**, Kwang-Sung Jun. International Conference on Artificial Intelligence and Statistics. 2025.
- **CSSP'18**: ***Blind Equalization in the Presence of Co-channel Interference Based on Higher-Order Statistics.*** Guohua Wang, **Kapilan Balagopalan**, Sirajudeen Gulam Razul, Shang-Kee Ting, Chong Meng Samson See. Circuits, Systems, and Signal Processing. 2018.